

GFPO baseline (Shrivastava et al., 2025). GFPO oversamples G responses, keeps a top- k subset S , and zeroes gradient from the dropped responses.

$$\begin{aligned}
 S &= \text{TopK}_m\{1, \dots, G\}, \\
 m_i &= \begin{cases} -L_i, & \text{shortest filter} \\ r_i / \max(L_i, 1), & \text{token-efficiency filter} \end{cases} \\
 \mu_S &= \frac{1}{|S|} \sum_{j \in S} r_j, & \sigma_S &= \sqrt{\frac{1}{|S|} \sum_{j \in S} (r_j - \mu_S)^2} \\
 A_i^{\text{GFPO}} &= \mathbf{1}[i \in S] \frac{r_i - \mu_S}{\sigma_S + \epsilon}.
 \end{aligned}$$

Our reproduction uses $G = 16$, $k = 8$, raw rewards unchanged, and dropped responses kept in batch with zero advantage.